

# Arpan Pathak

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📺 YouTube: [youtube.com/@ArpanPathak](https://www.youtube.com/@ArpanPathak) | Seattle-based · Local to NVIDIA Redmond (onsite)  
🇺🇸 Visa: Requires H1B Transfer sponsorship

## SUMMARY

Senior Software Engineer, **8 years: AI inference serving, backend microservices**, cloud-native infrastructure at **Amazon, Microsoft, Oracle, Razorpay**. Strong **Python, C++, Rust, CUDA**; **BERT inference at 5,000 QPS** (ONNX, INT8) on **Docker/Kubernetes** with CI/CD; passionate about **CV/DL for road safety** on **NVIDIA Jetson Orin** (TensorRT).

## PERSONAL PROJECTS

- 🚗 **Driving-CivicSense: Full-Stack Edge AI (Rust, CUDA), NVIDIA Jetson Orin Nano Super NX** | [github](#)
- **Full-stack data collection: 50,000 miles** of driving footage across all weather and terrain via vehicle camera sensors and **LIDAR**; simulation-driven synthetic data; privacy-first, on-device processing on edge neural chips.
  - **Multi-chip edge system: YOLOv8/v11 INT8 → TensorRT at ~12 ms** on **Jetson Orin Nano Super NX**; sensor data streamed to a **Raspberry Pi Zero 2 W UDP server**; validated on **Raspberry Pi 5** and **Orange Pi Zero**.
  - **Containerized VLM inference**: building edge VLM serving with **vLLM** and **TensorRT-LLM** in Docker containers on the Jetson.
  - **CUDA-Oxide (Rust-to-CUDA)**: contributing GPU kernels and documentation fixes (unsafe intrinsics wrapped in explicit unsafe blocks); pushing Jetson limits with idiomatic, memory-safe, fearlessly concurrent kernels.
  - **Research & safety standards**: paper on **Intersection Blockage Prediction** | [paper](#); studying ISO 26262. Books: **Seeing Machines · CUDA Kernels** | [site](#)

## EXPERIENCE

- Software Developer II, Microsoft, Redmond, WA** Jun 2025 – Aug 2026
- **Containerized Cloud Infrastructure**: zero-trust sovereign-cloud networking with eBPF + Cilium enforcing **data residency & SecNumCloud** compliance; Kubebuilder operators reconcile state across AKS clusters (France, Germany).
  - **Systems Performance**: owned the **C++/C# → Rust** migration of Microsoft Information Protection; eliminated use-after-free/buffer-overflow/null-safety bugs; ~ **40% lower P99, 2x throughput, 3x lower memory**, six-figure savings.
  - **Automation**: AI engine auto-generating Cilium policies and scanning misconfigurations; review **4 hours** → **near-real-time**, zero drift.  
↳ **TechStack: Rust, C++, Python, eBPF, XDP, Cilium, Kubernetes (AKS), Docker, CI/CD, Linux**
- Senior Software Engineer, Oracle Cloud Infrastructure, Seattle, WA** Oct 2024 – Mar 2025
- **Backend Services & SDKs**: owned the **Terraform provider** (developer SDK) for Autonomous Database, replacing legacy control-plane tooling; **\$1.2M/month** projected savings.
  - **High-Performance Systems**: lock-free data structures cutting contention in high-concurrency read-write paths; + **45,000 ops/s**.
  - **Security & Key Management**: integrated OCI Security Vault with automated key rotation across control-plane services.  
↳ **TechStack: Java, Go, Rust, REST/gRPC, Terraform, OCI**
- Software Development Engineer II, Amazon, Seattle, WA & Hyderabad, India** Mar 2021 – Oct 2024
- **Production Inference Serving**: scaled low-latency **BERT-based inference** to **5,000 QPS** (ONNX Runtime, INT8 quantization, latency tuning); validated throughput/latency; cut triage SLA **7 days** → **2 hours**.
  - **Real-Time ML Ranking**: owned end-to-end (training, serving, latency tuning) XGBoost + Deep Learning ensemble across **FreeVee, Twitch, MiniTV, Prime Video, Amazon Retail**; **5% ad CTR lift, \$5.4M/month revenue**.
  - **Event-Driven Microservices**: led a team of 3 building ad-delivery microservices at < **50 ms** via real-time CDC (Kotlin); REST/gRPC APIs.
  - **Distributed Systems**: designed a uniqueness-constraint indexing SDK (**two-phase commit, serializable isolation, 3K writes/s**).
  - **Data Platform**: built the **1-petabyte** warehouse/lake (AWS Glue, Spark) for financial reporting & pay computation.
  - **Telemetry & Automation**: real-time monitoring and automated controls cut deal over-delivery ~ **30%** and budget overspend **25%**.  
↳ **TechStack: Python, Kotlin, PyTorch, XGBoost, ONNX Runtime, Spark, Kafka, DynamoDB, Redshift, AWS**
- SDE, Razorpay, Bengaluru, India** Aug 2020 – Feb 2021
- **Backend Microservices**: P2P Unified Payments Gateway (Go, Protobuf/gRPC): **3M+ transactions/hour** peak, **99.99% uptime, 150ms P99**; customers: **Zomato, Swiggy, Zerodha, Groww** (AWS, GCP).
- Software Engineer, Mindfire, Bhubaneswar, India** Aug 2018 – Feb 2020
- Java/Spring full-stack tools; AR product-search microservice ( **10K+ daily queries** ).

## SKILLS

↳ **Languages**: Python, C++, Rust, Go, Java, Kotlin, C, Shell/Bash  
↳ **AI/ML Inference**: Deep Learning, Computer Vision, PyTorch, XGBoost, BERT, LLMs, ONNX Runtime, TensorRT (INT8), vLLM, SGLang, TensorRT-LLM, YOLOv8/v11, Edge Inference, Quantization, MLOps  
↳ **GPU & Parallel Computing**: CUDA, CUDA-Oxide (Rust-to-CUDA), SIMT, GPU Kernel Programming, NVIDIA Jetson Orin, HPC  
↳ **Systems Programming & Performance**: Lock-Free Data Structures, Concurrency, Memory Safety, Performance Engineering, Low-Latency Systems, Computer Architecture  
↳ **Networking**: TCP/IP, mTLS, HTTP/2/3, QUIC, eBPF, XDP, Cilium, Load Balancing  
↳ **Distributed Systems & Streaming**: Consensus, Replication, Sharding, Kafka, Spark, CDC, Microservices, gRPC, REST, Protobuf  
↳ **Cloud & Infra**: AWS (EC2, ECS, EKS, Lambda, Kinesis, EMR, SageMaker, CDK), Azure Kubernetes Service (AKS), Google Kubernetes Engine (GKE), Kubernetes, Docker, Helm, Terraform, Kubebuilder, Istio, CI/CD  
↳ **Observability**: Prometheus, Grafana, AWS CloudWatch  
↳ **Data & Storage**: DynamoDB, PostgreSQL, Redis, Amazon Redshift, Amazon S3

## EDUCATION

**B.Tech, Computer Science & Engineering**, RCC Institute of Information Technology, Kolkata | 2018